

OpenHive Architecture & Design

OpenHive is an open-source, self-hosted Apiary Management System designed to scale from a single backyard hive to commercial operations.

- Offline-first PWA
- Self-hosted with Docker
- Open API
- Privacy-first
- AI-assisted, not AI-dependent
- Modular architecture
- Authentication & Organizations
- Apiaries & Hives
- Boxes, Frames & Equipment
- Queen Management
- Inspections
- Health & Disease
- Varroa Tracking
- Feeding
- Honey & Wax Production
- Inventory
- Weather & Historical Weather
- Bloom Calendar & Forage Database
- Maps & GPS
- Photos & Video Almanac
- AI Image Review
- Sharing & Mentoring
- Reporting
- Public REST API

Frontend: React + TypeScript + Vite + Tailwind + shadcn/ui + Capacitor

Backend: FastAPI

Database: PostgreSQL

Cache: Redis (optional)

Storage: Local filesystem or S3-compatible

Deployment: Docker Compose / Kubernetes (optional)

Phase 1: Foundation

Phase 2: Hive Management

Phase 3: Inspection & Health

Phase 4: Weather & Bloom Intelligence

Phase 5: AI Assistant

Phase 6: Native Apps